PID Controller with Dynamics Compensation

LVLH Tracking, Lyapunov Stability and Pole Placement

Controller and tuning record of 3 October 2026

1. Model and errors

Let $\mathcal{I}$ be the inertial frame, $\mathcal{B}$ the body frame, and $\mathcal{D}$ the desired LVLH frame. The desired $+z_{\mathcal{D}}$ axis points towards the centre of Earth. Let $C_{\mathcal{BI}}$ be a DCM matrix that maps components from $\mathcal{I}$ to $\mathcal{B}$, and $C_{\mathcal{DI}}$ from $\mathcal{I}$ to $\mathcal{D}$. Then using properites of orthogonal matrices it is possible to obtain that $C_{\mathcal{BD}} = C_{\mathcal{BI}} C_{\mathcal{DI}}^{\mathsf{T}}$ maps components from $\mathcal{D}$ to $\mathcal{B}$.

Euler's equation is

$$J\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (J\boldsymbol{\omega}) = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{ext}}, \qquad J = J^{\mathsf{T}} > 0, \qquad \boldsymbol{\omega} = \boldsymbol{\omega}_{\mathcal{BI}}^{\mathcal{B}}.$$

The inertia matrix is constant and diagonally dominant. The control torque and disturbance torque are expressed in $\mathcal{B}$.

Let's define a unit quaternion that represents $C_{\mathcal{BD}}$

$$\boldsymbol{q}_e = \begin{bmatrix} \eta \\ \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}, \qquad \eta^2 + \varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 = 1.$$

The model uses the scalar-first convention.

The quaternions $\boldsymbol{q}_e$ and $-\boldsymbol{q}_e$ represent the same rotation. Therefore, when $\eta < 0$, we replace $\boldsymbol{q}_e$ by $-\boldsymbol{q}_e$, changing all four signs without changing the attitude. This selects $\eta = \cos(\theta/2) \geq 0$, where $\theta \in [0, \pi]$ is the attitude error angle. The proof stays in $\eta > 0$, where this representation is smooth. The boundary $\eta = 0$ corresponds to 180°; there, the condition $\eta \geq 0$ no longer distinguishes the two quaternion signs.

Let $\boldsymbol{\varepsilon}$ be the vector part of the error quaternion:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}.$$

For an error angle θ about a unit axis $\boldsymbol{n}$,

$$2\boldsymbol{\varepsilon} = 2\boldsymbol{n}\sin(\theta/2) \simeq \boldsymbol{n}\theta \qquad (\theta \to 0).$$

Thus $2\boldsymbol{\varepsilon}$ approximates the rotation-error vector used in the feedback.

Define the actual and reference angular velocities by

$$\boldsymbol{\omega} = \boldsymbol{\omega}_{\mathcal{BI}}^{\mathcal{B}}, \qquad \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} = C_{\mathcal{BD}}\boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{D}}.$$

The reference angular acceleration is

$$\boldsymbol{a} = C_{\mathcal{BD}} \frac{d\boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{D}}}{dt}.$$

Define the angular-velocity error as

$$\boxed{\begin{aligned} \boldsymbol{e}_\omega &= \boldsymbol{\omega}_{\mathcal{BI}}^{\mathcal{B}} - \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} \\ &= \boldsymbol{\omega}_{\mathcal{BI}}^{\mathcal{B}} - C_{\mathcal{BD}}\boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{D}} = \boldsymbol{\omega}_{\mathcal{BD}}^{\mathcal{B}}. \end{aligned}}$$

For a kinematically consistent reference, the transport theorem gives

$$\boldsymbol{\omega}_{\mathcal{DI}}^{\dot{\mathcal{B}}} = -\boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} + \boldsymbol{a}, \qquad \dot{\boldsymbol{e}_\omega} = \dot{\boldsymbol{\omega}} + \boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} - \boldsymbol{a}.$$

The exact error kinematics are

$$\begin{bmatrix} \dot{\eta} \\ \dot{\varepsilon}_1 \\ \dot{\varepsilon}_2 \\ \dot{\varepsilon}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -\varepsilon_1 & -\varepsilon_2 & -\varepsilon_3 \\ \eta & -\varepsilon_3 & \varepsilon_2 \\ \varepsilon_3 & \eta & -\varepsilon_1 \\ -\varepsilon_2 & \varepsilon_1 & \eta \end{bmatrix} \boldsymbol{e}_\omega.$$

Thus $\dot{\boldsymbol{\varepsilon}} = \frac{1}{2}\eta \boldsymbol{e}_\omega + \frac{1}{2}\boldsymbol{\varepsilon} \times \boldsymbol{e}_\omega$.

2. From PD to PID

Write the commanded torque as

$$\boldsymbol{\tau}_c = \boldsymbol{\omega} \times (J\boldsymbol{\omega}) + J\boldsymbol{v}.$$

With exact state feedback and no saturation, substitution into Euler's equation gives

$$J\dot{\boldsymbol{\omega}} = \underbrace{-\boldsymbol{\omega} \times (J\boldsymbol{\omega}) + \boldsymbol{\omega} \times (J\boldsymbol{\omega})}_{\mathbf{0}} + J\boldsymbol{v} + \boldsymbol{\tau}_{\text{ext}}.$$

Since J is invertible,

$$\dot{\boldsymbol{\omega}} = \boldsymbol{v} + J^{-1}\boldsymbol{\tau}_{\text{ext}}.$$

Choose the PD virtual input

$$\boldsymbol{v} = -\boldsymbol{e}_\omega \times \boldsymbol{\omega}^{\mathcal{B}}_{\mathcal{DI}} + \boldsymbol{a} - 2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega.$$

Substitute it into the rate-error derivative obtained from the transport theorem:

$$\begin{aligned}
\dot{\boldsymbol{e}}_\omega &= \dot{\boldsymbol{\omega}} + \boldsymbol{e}_\omega \times \boldsymbol{\omega}^{\mathcal{B}}_{\mathcal{DI}} - \boldsymbol{a} \\
&= \left(-\boldsymbol{e}_\omega \times \boldsymbol{\omega}^{\mathcal{B}}_{\mathcal{DI}} + \boldsymbol{a} - 2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega\right) + J^{-1}\boldsymbol{\tau}_{\text{ext}} + \boldsymbol{e}_\omega \times \boldsymbol{\omega}^{\mathcal{B}}_{\mathcal{DI}} - \boldsymbol{a} \\
&= \underbrace{-\boldsymbol{e}_\omega \times \boldsymbol{\omega}^{\mathcal{B}}_{\mathcal{DI}} + \boldsymbol{e}_\omega \times \boldsymbol{\omega}^{\mathcal{B}}_{\mathcal{DI}}}_{\mathbf{0}} + \underbrace{\boldsymbol{a} - \boldsymbol{a}}_{\mathbf{0}} - 2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega + J^{-1}\boldsymbol{\tau}_{\text{ext}}.
\end{aligned}$$

Thus the gyroscopic and reference terms cancel exactly, leaving

$$\boxed{\dot{\boldsymbol{e}}_\omega = -2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega + J^{-1}\boldsymbol{\tau}_{\text{ext}}.}$$

The local second order equation. Near $\boldsymbol{\varepsilon} = \mathbf{0}$, $\boldsymbol{e}_\omega = \mathbf{0}$, we have $\eta \simeq 1$ and the product $\boldsymbol{\varepsilon} \times \boldsymbol{e}_\omega$ is of second order. The exact kinematics therefore give

$$\dot{\boldsymbol{\varepsilon}} = \tfrac{1}{2}\eta\boldsymbol{e}_\omega + \tfrac{1}{2}\boldsymbol{\varepsilon} \times \boldsymbol{e}_\omega \quad \Longrightarrow \quad 2\dot{\boldsymbol{\varepsilon}} \simeq \boldsymbol{e}_\omega.$$

In the linearised model, substitute $\boldsymbol{e}_\omega = 2\dot{\boldsymbol{\varepsilon}}$ and $\dot{\boldsymbol{e}}_\omega = 2\ddot{\boldsymbol{\varepsilon}}$ into the compensated dynamics:

$$2\ddot{\boldsymbol{\varepsilon}} = -2k_p\boldsymbol{\varepsilon} - 2k_d\dot{\boldsymbol{\varepsilon}} + J^{-1}\boldsymbol{\tau}_{\text{ext}}.$$

Dividing by two gives

$$\boxed{\ddot{\boldsymbol{\varepsilon}} + k_d\dot{\boldsymbol{\varepsilon}} + k_p\boldsymbol{\varepsilon} = \tfrac{1}{2}J^{-1}\boldsymbol{\tau}_{\text{ext}}.}$$

For zero disturbance, each component satisfies $\ddot{\varepsilon}_j + k_d\dot{\varepsilon}_j + k_p\varepsilon_j = 0$. This is a linear second order model on each axis.

Adding integral action. The integral term is introduced to compensate an unknown external torque and remove the resulting steady attitude error. It accumulates the attitude error and produces a torque $-J\boldsymbol{\xi}$, without requiring the disturbance as an input. For a disturbance constant in $\mathcal{B}$, this torque can balance the disturbance at zero attitude error, as shown below.

Let's define the integral state

$$\boldsymbol{\xi} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} \in \mathbb{R}^3, \qquad \dot{\boldsymbol{\xi}} = 2k_i\boldsymbol{\varepsilon}.$$

Bold symbols denote vectors; unbold symbols with an axis index denote their scalar components. In particular, $\dot{\xi}_j = 2k_i\varepsilon_j$ for $j = 1, 2, 3$. Then the virtual control can be defined as

$$\boldsymbol{v} = -\boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} + \boldsymbol{a} - 2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega - \boldsymbol{\xi}.$$

Hence the PID torque demand is

$$\boldsymbol{\tau}_c = \boldsymbol{\omega} \times (J\boldsymbol{\omega}) + J\big(-\boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} + \boldsymbol{a} - 2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega - \boldsymbol{\xi}\big).$$

Here k_p, k_d, k_i are positive scalars, equal on all axes.

With exact state feedback and no saturation, substitute this torque into Euler's equation:

$$\begin{aligned} J\dot{\boldsymbol{\omega}} = &\underbrace{-\boldsymbol{\omega} \times (J\boldsymbol{\omega}) + \boldsymbol{\omega} \times (J\boldsymbol{\omega})}_{\mathbf{0}} \\ &+ J\big(-\boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} + \boldsymbol{a} - 2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega - \boldsymbol{\xi}\big) + \boldsymbol{\tau}_{\text{ext}}. \end{aligned}$$

Multiplying by J^{-1} and substituting into the rate-error derivative gives

$$\begin{aligned} \dot{\boldsymbol{e}_\omega} &= \dot{\boldsymbol{\omega}} + \boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} - \boldsymbol{a} \\ &= \big(-\boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} + \boldsymbol{a} - 2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega - \boldsymbol{\xi}\big) + J^{-1}\boldsymbol{\tau}_{\text{ext}} + \boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} - \boldsymbol{a} \\ &= \underbrace{-\boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}} + \boldsymbol{e}_\omega \times \boldsymbol{\omega}_{\mathcal{DI}}^{\mathcal{B}}}_{\mathbf{0}} + \underbrace{\boldsymbol{a} - \boldsymbol{a}}_{\mathbf{0}} - 2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega - \boldsymbol{\xi} + J^{-1}\boldsymbol{\tau}_{\text{ext}}. \end{aligned}$$

Thus the exact closed-loop equations are

$$\boxed{\begin{aligned} \dot{\boldsymbol{\varepsilon}} &= \tfrac{1}{2}\eta\boldsymbol{e}_\omega + \tfrac{1}{2}\boldsymbol{\varepsilon} \times \boldsymbol{e}_\omega, \\ \dot{\boldsymbol{e}_\omega} &= -2k_p\boldsymbol{\varepsilon} - k_d\boldsymbol{e}_\omega - \boldsymbol{\xi} + J^{-1}\boldsymbol{\tau}_{\text{ext}}, \\ \dot{\boldsymbol{\xi}} &= 2k_i\boldsymbol{\varepsilon}. \end{aligned}}$$

The local third order equation. We first write the equations in vector form, with $\boldsymbol{\varepsilon}, \boldsymbol{e}_\omega, \boldsymbol{\xi} \in \mathbb{R}^3$ and scalar gains k_p, k_d, k_i. As previously mentioned, near the desired attitude the linearised kinematics give $\boldsymbol{e}_\omega = 2\dot{\boldsymbol{\varepsilon}}$ and $\dot{\boldsymbol{e}_\omega} = 2\ddot{\boldsymbol{\varepsilon}}$. Substitution gives

$$2\ddot{\boldsymbol{\varepsilon}} = -2k_p\boldsymbol{\varepsilon} - 2k_d\dot{\boldsymbol{\varepsilon}} - \boldsymbol{\xi} + J^{-1}\boldsymbol{\tau}_{\text{ext}}.$$

Dividing by two and collecting the attitude terms:

$$\ddot{\boldsymbol{\varepsilon}} + k_d\dot{\boldsymbol{\varepsilon}} + k_p\boldsymbol{\varepsilon} = -\tfrac{1}{2}\boldsymbol{\xi} + \tfrac{1}{2}J^{-1}\boldsymbol{\tau}_{\text{ext}}.$$

Differentiating the first equation gives

$$\frac{d^3\boldsymbol{\varepsilon}}{dt^3} + k_d\ddot{\boldsymbol{\varepsilon}} + k_p\dot{\boldsymbol{\varepsilon}} = -\tfrac{1}{2}\dot{\boldsymbol{\xi}} + \tfrac{1}{2}J^{-1}\dot{\boldsymbol{\tau}}_{\text{ext}}.$$

Substitute $\dot{\boldsymbol{\xi}} = 2k_i\boldsymbol{\varepsilon}$:

$$\frac{d^3\boldsymbol{\varepsilon}}{dt^3} + k_d\ddot{\boldsymbol{\varepsilon}} + k_p\dot{\boldsymbol{\varepsilon}} = -\underbrace{\tfrac{1}{2}(2k_i\boldsymbol{\varepsilon})}_{k_i\boldsymbol{\varepsilon}} + \tfrac{1}{2}J^{-1}\dot{\boldsymbol{\tau}}_{\text{ext}}.$$

Moving the integral feedback term to the left gives

$$\boxed{\frac{d^3\boldsymbol{\varepsilon}}{dt^3} + k_d\ddot{\boldsymbol{\varepsilon}} + k_p\dot{\boldsymbol{\varepsilon}} + k_i\boldsymbol{\varepsilon} = \tfrac{1}{2}J^{-1}\dot{\boldsymbol{\tau}}_{\text{ext}}.}$$

Scalar equations for three axes have the same coefficients but may have different initial conditions. Their independence holds in the local linearised model.

On $\eta > 0$, nadir pointing equilibrium is defined as

$$\boxed{\boldsymbol{q}_e^* = \begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix}^{\mathsf{T}}, \qquad \boldsymbol{e}_\omega^* = \mathbf{0}, \qquad \boldsymbol{\xi}^* = J^{-1}\boldsymbol{\tau}_{\text{ext},0}.}$$

At this equilibrium, the integral torque and the external torque cancel:

$$-J\boldsymbol{\xi}^* + \boldsymbol{\tau}_{\text{ext}} = -JJ^{-1}\boldsymbol{\tau}_{\text{ext}} + \boldsymbol{\tau}_{\text{ext}} = \mathbf{0}.$$

When $\boldsymbol{\varepsilon} = \mathbf{0}$, we have $\dot{\boldsymbol{\xi}} = \mathbf{0}$. The integral state retains its accumulated value and continues to supply the balancing torque. Thus nonzero attitude error is no longer needed to sustain this torque.

3. Preparation for the stability proof

Assume exact state feedback and inertia, a kinematically consistent reference, continuous control without saturation, and a disturbance $\boldsymbol{\tau}_{\mathrm{ext},0}$ constant in $\mathcal{B}$.

To simplify the analysis, choose a fixed frequency scale $\omega_0 > 0$, in s^{-1}, and introduce dimensionless time:

$$\vartheta = \omega_0 t.$$

A prime will denote differentiation with respect to this time, so

$$\frac{d}{d\vartheta} = \frac{1}{\omega_0}\frac{d}{dt}.$$

Next, scale the angular rate error:

$$\boldsymbol{u} = \frac{\boldsymbol{e}_\omega}{\omega_0}.$$

The integral state has the equilibrium value $\boldsymbol{\xi}^* = J^{-1}\boldsymbol{\tau}_{\mathrm{ext},0}$. Subtract this value and scale the result to define the integral error:

$$\boldsymbol{z} = \frac{\boldsymbol{\xi} - J^{-1}\boldsymbol{\tau}_{\mathrm{ext},0}}{\omega_0^2}.$$

Both errors are dimensionless and vanish at the desired equilibrium. Collect them with the attitude error into one state vector:

$$\boldsymbol{x} = \begin{bmatrix} 2\boldsymbol{\varepsilon} \\ \boldsymbol{u} \\ \boldsymbol{z} \end{bmatrix} \in \mathbb{R}^9.$$

Thus the equilibrium is $\boldsymbol{x} = \mathbf{0}$. Finally, define the dimensionless gains that appear when the dynamics are written in the new time:

$$\kappa_p = \frac{k_p}{\omega_0^2}, \qquad \kappa_d = \frac{k_d}{\omega_0}, \qquad \kappa_i = \frac{k_i}{\omega_0^3}.$$

First, the unit quaternion constraint gives $\eta^2 + \|\boldsymbol{\varepsilon}\|^2 = 1$. On the branch $\eta > 0$,

$$\eta = \sqrt{1 - \|\boldsymbol{\varepsilon}\|^2}, \qquad \|\boldsymbol{\varepsilon}\| < 1.$$

Next, substitute the scaled states and gains into the error equations:

$$\begin{aligned} (2\boldsymbol{\varepsilon})' &= \eta\boldsymbol{u} + \boldsymbol{\varepsilon} \times \boldsymbol{u}, \\ \boldsymbol{u}' &= -\kappa_p(2\boldsymbol{\varepsilon}) - \kappa_d\boldsymbol{u} - \boldsymbol{z}, \\ \boldsymbol{z}' &= \kappa_i(2\boldsymbol{\varepsilon}). \end{aligned}$$

To separate the linear terms, write $\eta\boldsymbol{u} = \boldsymbol{u} + (\eta - 1)\boldsymbol{u}$. The linear part on one axis therefore has the matrix

$$A_0 = \begin{bmatrix} 0 & 1 & 0 \\ -\kappa_p & -\kappa_d & -1 \\ \kappa_i & 0 & 0 \end{bmatrix}.$$

For the state $\boldsymbol{x} = [(2\boldsymbol{\varepsilon})^\mathsf{T}, \boldsymbol{u}^\mathsf{T}, \boldsymbol{z}^\mathsf{T}]^\mathsf{T}$, the full matrix is

$$\mathcal{A} = \begin{bmatrix} 0_{3\times3} & I_3 & 0_{3\times3} \\ -\kappa_p I_3 & -\kappa_d I_3 & -I_3 \\ \kappa_i I_3 & 0_{3\times3} & 0_{3\times3} \end{bmatrix} \in \mathbb{R}^{9\times9}.$$

Each block is 3×3; the three block rows give the equations for $2\boldsymbol{\varepsilon}$, $\boldsymbol{u}$ and $\boldsymbol{z}$, respectively. The remaining nonlinear terms are collected in

$$\boldsymbol{F}(\boldsymbol{x}) = \begin{bmatrix} (\eta - 1)\boldsymbol{u} + \boldsymbol{\varepsilon} \times \boldsymbol{u} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}.$$

Thus the exact scaled dynamics take the form

$$\boxed{\boldsymbol{x}' = \mathcal{A}\boldsymbol{x} + \boldsymbol{F}(\boldsymbol{x}).}$$

No nonlinear term has been discarded; the linearisation alone is $\boldsymbol{x}' = \mathcal{A}\boldsymbol{x}$.

Recall the linearised equation for one axis under a constant disturbance:

$$\frac{d^3\varepsilon_j}{dt^3} + k_d \frac{d^2\varepsilon_j}{dt^2} + k_p \frac{d\varepsilon_j}{dt} + k_i \varepsilon_j = 0, \qquad j = 1, 2, 3.$$

Applying the Laplace transform with respect to t to this equation gives the characteristic polynomial

$$p(s) = s^3 + k_d s^2 + k_p s + k_i.$$

We use the Routh–Hurwitz criterion to determine when all roots of $p(s)$ have negative real parts. The first column of the Routh table is

$$1, \qquad k_d, \qquad \frac{k_d k_p - k_i}{k_d}, \qquad k_i.$$

Since the leading coefficient is positive, all entries in this column must be strictly positive. Thus $k_d > 0$, $k_i > 0$ and $k_d k_p > k_i$, which give

$$\boxed{k_p > 0, \qquad k_d > 0, \qquad 0 < k_i < k_p k_d.}$$

To relate this result directly to A_0, compute its characteristic polynomial:

$$\det(\lambda I_3 - A_0) = \begin{vmatrix} \lambda & -1 & 0 \\ \kappa_p & \lambda + \kappa_d & 1 \\ -\kappa_i & 0 & \lambda \end{vmatrix} = \lambda^3 + \kappa_d \lambda^2 + \kappa_p \lambda + \kappa_i.$$

The eigenvalues $\lambda_1, \lambda_2, \lambda_3$ are the roots of this polynomial. By the definitions of the scaled gains,

$$\det(\lambda I_3 - A_0) = \frac{p(\omega_0 \lambda)}{\omega_0^3}.$$

Thus, if s_j is a root of $p(s)$, then $\lambda_j = s_j/\omega_0$. Since $\omega_0 > 0$, the Routh–Hurwitz condition above gives $\operatorname{Re} \lambda_j < 0$: A_0 is Hurwitz. Each eigenvalue occurs three times in $\mathcal{A}$, so $\mathcal{A}$ is also Hurwitz.

Thus, for every choice of gains satisfying the strict condition,

$$\boxed{k_p > 0, \quad k_d > 0, \quad 0 < k_i < k_p k_d, \quad \omega_0 > 0 \quad \Longrightarrow \quad \operatorname{Re} \lambda_j(A_0) < 0, \quad j = 1, 2, 3.}$$

We use this general result below to construct a Lyapunov function.

4. A Lyapunov function for the nonlinear error system

Assume $k_p > 0$, $k_d > 0$ and $0 < k_i < k_p k_d$. By the previous section, all eigenvalues of A_0 have negative real parts, hence A_0 is Hurwitz. We use this property to construct a Lyapunov function, then check its derivative along the full nonlinear dynamics.

Choosing the matrix. We seek a quadratic function $V(\boldsymbol{x}) = \boldsymbol{x}^{\mathsf{T}}\mathcal{P}\boldsymbol{x}$ with $\mathcal{P}$ symmetric positive definite. For the linear part alone, its derivative would be

$$V' = \boldsymbol{x}^{\mathsf{T}}(\mathcal{A}^{\mathsf{T}}\mathcal{P} + \mathcal{P}\mathcal{A})\boldsymbol{x}.$$

Thus we choose the matrix so that the expression in parentheses equals $-I_9$. Since the three axes have the same linear dynamics, first solve

$$A_0^{\mathsf{T}}P_0 + P_0A_0 = -I_3.$$

Because A_0 is Hurwitz, a symmetric positive definite solution is

$$P_0 = \int_0^\infty e^{A_0^{\mathsf{T}}r}e^{A_0 r}\,dr.$$

Here r is only an integration variable. The matrix exponential decays as $r \to \infty$, so the integral converges. For any nonzero $\boldsymbol{b} \in \mathbb{R}^3$,

$$\boldsymbol{b}^{\mathsf{T}}P_0\boldsymbol{b} = \int_0^\infty \left\|e^{A_0 r}\boldsymbol{b}\right\|^2 dr > 0.$$

This proves positive definiteness. Moreover,

$$A_0^{\mathsf{T}}P_0 + P_0A_0 = \int_0^\infty \frac{d}{dr}\left(e^{A_0^{\mathsf{T}}r}e^{A_0 r}\right) dr = 0_{3\times 3} - I_3 = -I_3.$$

Thus P_0 satisfies the required equation; see [1].

Writing $P_0 = (p_{ij})$, define

$$\mathcal{P} = \begin{bmatrix} p_{11}I_3 & p_{12}I_3 & p_{13}I_3 \\ p_{12}I_3 & p_{22}I_3 & p_{23}I_3 \\ p_{13}I_3 & p_{23}I_3 & p_{33}I_3 \end{bmatrix} \in \mathbb{R}^{9\times 9}.$$

It is symmetric positive definite and, by block multiplication, satisfies

$$\mathcal{A}^{\mathsf{T}}\mathcal{P} + \mathcal{P}\mathcal{A} = -I_9.$$

The Lyapunov function. Choose

$$\boxed{V(\boldsymbol{x}) = \boldsymbol{x}^{\mathsf{T}}\mathcal{P}\boldsymbol{x}.}$$

For $\boldsymbol{x} = [(2\boldsymbol{\varepsilon})^{\mathsf{T}}, \boldsymbol{u}^{\mathsf{T}}, \boldsymbol{z}^{\mathsf{T}}]^{\mathsf{T}}$, this gives

$$\begin{aligned} V = {} & 4p_{11}\|\boldsymbol{\varepsilon}\|^2 + p_{22}\|\boldsymbol{u}\|^2 + p_{33}\|\boldsymbol{z}\|^2 \\ & + 4p_{12}\boldsymbol{\varepsilon}^{\mathsf{T}}\boldsymbol{u} + 4p_{13}\boldsymbol{\varepsilon}^{\mathsf{T}}\boldsymbol{z} + 2p_{23}\boldsymbol{u}^{\mathsf{T}}\boldsymbol{z}. \end{aligned}$$

Let $m = \lambda_{\min}(P_0) > 0$ and $M = \lambda_{\max}(P_0)$. These are also the smallest and largest eigenvalues of $\mathcal{P}$. Therefore

$$m\|\boldsymbol{x}\|^2 \le V(\boldsymbol{x}) \le M\|\boldsymbol{x}\|^2.$$

Hence $V(\boldsymbol{0}) = 0$ and $V(\boldsymbol{x}) > 0$ for $\boldsymbol{x} \neq \boldsymbol{0}$.

Derivative along the nonlinear dynamics. Substitute $\boldsymbol{x}' = \mathcal{A}\boldsymbol{x} + \boldsymbol{F}(\boldsymbol{x})$:

$$\begin{aligned} V' &= (\boldsymbol{x}')^{\mathsf{T}}\mathcal{P}\boldsymbol{x} + \boldsymbol{x}^{\mathsf{T}}\mathcal{P}\boldsymbol{x}' \\ &= \boldsymbol{x}^{\mathsf{T}}(\mathcal{A}^{\mathsf{T}}\mathcal{P} + \mathcal{P}\mathcal{A})\boldsymbol{x} + 2\boldsymbol{x}^{\mathsf{T}}\mathcal{P}\boldsymbol{F}(\boldsymbol{x}) \\ &= -\left\|\boldsymbol{x}\right\|^2 + 2\boldsymbol{x}^{\mathsf{T}}\mathcal{P}\boldsymbol{F}(\boldsymbol{x}). \end{aligned}$$

The first term is negative away from zero. The second has no fixed sign, so we must bound its magnitude.

Bounding the nonlinear term. Recall that only the first block of $\boldsymbol{F}$ is nonzero:

$$\boldsymbol{F}(\boldsymbol{x}) = \begin{bmatrix} (\eta - 1)\boldsymbol{u} + \boldsymbol{\varepsilon} \times \boldsymbol{u} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}.$$

On the branch $\eta > 0$, the unit-quaternion constraint gives

$$1 - \eta = \frac{\left\|\boldsymbol{\varepsilon}\right\|^2}{1 + \eta} \leq \left\|\boldsymbol{\varepsilon}\right\|^2.$$

Together with $\left\|\boldsymbol{\varepsilon} \times \boldsymbol{u}\right\| \leq \left\|\boldsymbol{\varepsilon}\right\| \left\|\boldsymbol{u}\right\|$, this implies

$$\left\|\boldsymbol{F}(\boldsymbol{x})\right\| \leq \left\|\boldsymbol{\varepsilon}\right\|^2 \left\|\boldsymbol{u}\right\| + \left\|\boldsymbol{\varepsilon}\right\| \left\|\boldsymbol{u}\right\|.$$

Since $\boldsymbol{x}$ contains $2\boldsymbol{\varepsilon}$ and $\boldsymbol{u}$,

$$\left\|\boldsymbol{\varepsilon}\right\| \leq \tfrac{1}{2}\left\|\boldsymbol{x}\right\|, \qquad \left\|\boldsymbol{u}\right\| \leq \left\|\boldsymbol{x}\right\|.$$

For $\left\|\boldsymbol{x}\right\| \leq 1$, we therefore obtain

$$\left\|\boldsymbol{F}(\boldsymbol{x})\right\| \leq \tfrac{1}{4}\left\|\boldsymbol{x}\right\|^3 + \tfrac{1}{2}\left\|\boldsymbol{x}\right\|^2 \leq \tfrac{3}{4}\left\|\boldsymbol{x}\right\|^2.$$

Using the Euclidean induced matrix norm $\left\|\mathcal{P}\right\| = M$,

$$\left|2\boldsymbol{x}^{\mathsf{T}}\mathcal{P}\boldsymbol{F}(\boldsymbol{x})\right| \leq 2M\left\|\boldsymbol{x}\right\|\left\|\boldsymbol{F}(\boldsymbol{x})\right\| \leq \tfrac{3}{2}M\left\|\boldsymbol{x}\right\|^3.$$

Substituting this bound into V' gives

$$V' \leq -\left\|\boldsymbol{x}\right\|^2 + \tfrac{3}{2}M\left\|\boldsymbol{x}\right\|^3 = -\left(1 - \tfrac{3}{2}M\left\|\boldsymbol{x}\right\|\right)\left\|\boldsymbol{x}\right\|^2.$$

A neighbourhood where the derivative is negative. Choose

$$\rho = \min\left\{1, \frac{1}{3M}\right\}.$$

Then $\left\|\boldsymbol{x}\right\| \leq \rho$ ensures both the bound above and $\frac{3}{2}M\left\|\boldsymbol{x}\right\| \leq \frac{1}{2}$. Hence

$$\boxed{V' \leq -\tfrac{1}{2}\left\|\boldsymbol{x}\right\|^2 < 0 \qquad \text{for } 0 < \left\|\boldsymbol{x}\right\| \leq \rho.}$$

Thus V decreases in this neighbourhood for the full nonlinear system. The next section selects an invariant subset of this neighbourhood and proves convergence to the equilibrium.

5. Invariance, convergence and the numerical certificate

For $0 < c < m\rho^2$, define

$$\Omega_c = \{\boldsymbol{x} \in \mathbb{R}^9 : V(\boldsymbol{x}) \le c\}.$$

Since $m\,\|\boldsymbol{x}\|^2 \le V(\boldsymbol{x})$, every point of Ω_c satisfies $\|\boldsymbol{x}\| < \rho$. The set is compact and lies strictly inside the smooth quaternion branch. While a trajectory is in this set, $V' \le 0$. It cannot leave through $V = c$, where $V' < 0$. Hence Ω_c is positively invariant and solutions starting there exist for all future time.

Moreover, $V \le M\,\|\boldsymbol{x}\|^2$ implies

$$V' \le -\frac{V}{2M}, \qquad V(t) \le V(0)\exp\!\left(-\frac{\omega_0 t}{2M}\right).$$

Therefore

$$\|\boldsymbol{x}(t)\| \le \sqrt{\frac{M}{m}}\,\exp\!\left(-\frac{\omega_0 t}{4M}\right)\|\boldsymbol{x}(0)\|.$$

Thus $\boldsymbol{x} = \mathbf{0}$ is locally exponentially stable, and in particular locally asymptotically stable in the sense of Lyapunov. Returning to the original variables,

$$\boxed{\boldsymbol{q}_e(t) \longrightarrow \begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix}^{\mathsf{T}}, \qquad \boldsymbol{e}_\omega(t) \longrightarrow \mathbf{0}, \qquad \boldsymbol{\xi}(t) \longrightarrow J^{-1}\boldsymbol{\tau}_{\mathrm{ext},0}.}$$

For the selected gains and $\omega_0 = 0.06\ \mathrm{s}^{-1}$,

$$(\kappa_p, \kappa_d, \kappa_i) = (1.09,\ 2.4375,\ 0.0375),$$

and the Lyapunov equation gives, with rounded entries,

$$P_0 \simeq \begin{bmatrix}1.690365 & 0.514971 & 1.635148\\ 0.514971 & 0.416398 & 0.500000\\ 1.635148 & 0.500000 & 28.265885\end{bmatrix}.$$

Its eigenvalues are approximately 0.234063, 1.762399, 28.376187. The unrounded solution gives

$$\rho \simeq 0.01174694, \qquad m\rho^2 \simeq 3.22985 \cdot 10^{-5}.$$

For example, $c = 3.2 \cdot 10^{-5}$ defines a sufficient invariant set. The initial condition is tested on the *combined* scaled state $(2\boldsymbol{\varepsilon}, \boldsymbol{e}_\omega/\omega_0, (\boldsymbol{\xi} - J^{-1}\boldsymbol{\tau}_{\mathrm{ext},0})/\omega_0^2)$, not on attitude alone. This certificate is conservative. It neither gives the full region of attraction nor certifies the simulated acquisitions from 30° or 90°.

6. Saturation, implementation and limits of the proof

The proof establishes local stability for continuous control without saturation, with exact state feedback, exact compensation and a constant external torque. It does not cover sampling, estimation errors, varying disturbances or resets.

7. Tuning by pole placement

The tuning uses the same local polynomial as the stability calculation. Choose $\omega_n > 0$, $\zeta > 0$, $p_I > 0$ and require

$$p(s) = (s + p_I)(s^2 + 2\zeta\omega_n s + \omega_n^2).$$

Expansion and comparison with $s^3 + k_d s^2 + k_p s + k_i$ give

$$\boxed{\begin{aligned} k_d &= 2\zeta\omega_n + p_I, \\ k_p &= \omega_n^2 + 2\zeta\omega_n p_I, \\ k_i &= \omega_n^2 p_I. \end{aligned}}$$

This is pole placement for the state consisting of attitude error, rate error and integral state. It was performed by coefficient matching. The general design principle is described in [2].

The quadratic factor sets two nominal attitude poles. The remaining pole is $-p_I$. Decreasing p_I makes this mode slower, increasing it raises k_i and also changes the required P and D gains. Tuning k_i alone while retaining arbitrary P and D gains does not preserve the chosen poles.

The selected values are

$$\omega_n = 0.06 \text{ rad/s}, \qquad \zeta = 1.2, \qquad p_I = 0.00225 \text{ s}^{-1},$$

$$\boxed{k_p = 0.003924 \text{ s}^{-2}, \quad k_d = 0.14625 \text{ s}^{-1}, \quad k_i = 8.1 \cdot 10^{-6} \text{ s}^{-3}.}$$

The poles are -0.1117995, -0.0322005, -0.00225 s^{-1}. Also, $k_p k_d = 5.73885 \cdot 10^{-4}$ s^{-3} $> k_i$.

For a locally held acceleration command and forward-Euler integral, the sampled one-axis matrix in physical coordinates $(2\varepsilon_j, e_{\omega,j}, \xi_j - [J^{-1}\boldsymbol{\tau}_{\text{ext},0}]_j)$ is

$$A_d = \begin{bmatrix} 1 - \frac{1}{2}k_p T_s^2 & T_s - \frac{1}{2}k_d T_s^2 & -\frac{1}{2}T_s^2 \\ -k_p T_s & 1 - k_d T_s & -T_s \\ k_i T_s & 0 & 1 \end{bmatrix}.$$

For $T_s = 0.1$ s its eigenvalues are approximately 0.9887925, 0.9967879, 0.9997750, all inside the unit circle. This checks the sampled local model. The anti-windup time $T_{aw} = 150$ s was selected through simulation with clipping.

8. Recorded tuning trials

First, 120 combinations were screened in a one axis model with exact angle kinematics, sampled PID, and torque clipping. The grid was

$$\omega_n \in \{0.02, 0.03, 0.04, 0.05, 0.06\}, \quad p_I/\omega_n \in \{0.01, 0.025, 0.05, 0.1\},$$
$$\zeta \in \{1, 1.2\}, \qquad\qquad T_{aw} \in \{10, 50, 150\} \text{ s}.$$

Each combination used 30° and 90° acquisitions without disturbance, and a 5° case with synthetic torque $[0.9 + 0.25\sin(2\pi t/5700)]$ μN m, over 1800 s.

All tuning runs below use a 30° initial rotation about $[1\ 1\ 1]^{\mathsf{T}}/\sqrt{3}$, zero initial rate error and 1200 s duration. Environmental disturbances, sensor seeds, reference and actuator limits were held fixed.

Trial	k_p [s^{-2}]	k_d [s^{-1}]	k_i [s^{-3}]	T_{aw} [s]
T00	0.000400	0.04000	$2.00 \cdot 10^{-6}$	10
T01	0.000990	0.06150	$1.35 \cdot 10^{-6}$	10
T02	0.003000	0.10500	$1.25 \cdot 10^{-5}$	50
T03	0.004320	0.12600	$2.16 \cdot 10^{-5}$	50
T04	0.003816	0.14550	$5.40 \cdot 10^{-6}$	150
T05	0.004032	0.14700	$1.08 \cdot 10^{-5}$	150
T06	0.003924	0.14625	$8.10 \cdot 10^{-6}$	150

Let R be the maximum attitude error after first entering 1°, t_1 the time after which the error stays below 1°, and E_{60} the RMS over the final 60 s. These use true attitude for evaluation. The declared targets were $R \le 2°$, $t_1 \le 600$ s, $E_{60} \le 0.25°$.

Trial	R [°]	t_1 [s]	E_{60} [°]	Decision
T00	8.851	602.9	0.186	Reference; large rebound
T01	2.350	1028.7	0.764	Reject; all targets missed
T02	3.316	412.0	0.052	Reject; rebound
T03	2.964	335.6	0.034	Reject; rebound
T04	1.402	423.2	0.325	Reject; final RMS
T05	2.400	503.8	0.142	Reject; rebound
T06	1.955	507.2	0.227	**Select**

T01–T03 used $\zeta = 1$. T04–T06 used $\omega_n = 0.06$, $\zeta = 1.2$, with $p_I = 0.0015$, 0.003, then 0.00225 s^{-1}. T06 satisfies the declared compromise between rebound and holding error.

9. Validation and reproducibility

The selected gains were then held fixed for three further full-plant runs. The order V01, V02, V03 is the execution order. The two 90° cases used zero initial rate error. V02 used $\boldsymbol{e}_\omega(0) = [0.1 \ -0.05 \ 0.03]^\mathsf{T}$ °/s.

Run	Initial attitude	Duration [s]	t_1 [s]	E_{60} [°]	Clipping [%]
V01	90°, body x	1800	194.4	0.073	2.54
V02	5°, body z	3000	56.6	0.570	0.00
V03	90°, $[1\ 1\ 1]^\mathsf{T}/\sqrt{3}$	1800	822.8	0.169	1.73

Clipping is the fraction of active samples where the torque demand reaches beyond a component limit. Both 90° runs met $t_1 \leq 900$ s and $E_{60} \leq 0.3°$. All recorded integral states were finite and applied torques respected the 20 μN m per axis limit.

In V02, the final window RMS values were 0.570° for true tracking, 0.091° for estimated tracking, and 0.617° for MEKF estimation error. The integral remained bounded; the true pointing error was dominated by estimation error in the final interval.

Compared with T00, T06 reduced rebound but increased the 1200 s tail RMS from 0.186° to 0.227°. The tail RMS norm of applied torque increased from 0.769 to 2.333 μN m, consistent with the larger rate feedback gain. Four controller tests passed after saving the selected configuration.

References

[1] H. K. Khalil, *Nonlinear Control, Lecture 3: Stability of Equilibrium Points*, pp. 13-17, 22. Michigan State University lecture notes.

[2] MathWorks, *Pole Placement*. Control System Toolbox documentation.

[3] MathWorks, *Anti-Windup Control Using PID Controller Block*. Simulink documentation.